

When Female Doctors Fly In: Access to Female Physicians in Primary Care

Martina Repetto*

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Abstract

In many rural communities, a woman who wants to see a female doctor has to leave town. I study how this barrier shapes women's primary care, using a program that periodically sends a visiting female general practitioner to rural Australian towns and linked administrative records covering every subsidised consultation, prescription, and test. Two designs answer two questions. Comparing the same woman's encounters with the visiting female physician and with local male providers, genitourinary or hormone medications are between three and four times as common among her prescription days with the female physician, and cervical screening is more than ten times as common among her service days. Cardiovascular and mental-health prescribing do not differ, and encounters with other non-local male providers show no similar pattern. Around the program's entry into a town, prescribing does not change on average, but women who come to use the service become more likely to receive gender-specific prescriptions each month without seeing a doctor more often. The pattern is consistent both with women being more comfortable raising hormonal and gynaecological concerns with a female doctor and with differences in practice style. The findings suggest that the gender composition of the local workforce, and not only its size, shapes which health needs are met.

JEL Codes: I11, I12, I14, I18, J16

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*Department of Economics, University of Melbourne. Email: martina.repetto@student.unimelb.edu.au. I am grateful to the Royal Flying Doctor Service Victoria for providing institutional details, program data, and patient survey results. This research uses data from the Person-Level Integrated Data Asset (PLIDA) accessed through the ABS DataLab.

1 Introduction

In many rural communities, a woman who wants to see a female doctor has to leave town. Rural areas face persistent shortages of primary care physicians ([OECD and European Commission, 2024](#)), and the growth in the share of female physicians over recent decades has been concentrated in cities ([Ono et al., 2014](#); [Lally et al., 2026](#)). Many rural towns therefore have no female doctor, regardless of how many doctors practise there. Access to primary care in these communities is limited not only in how many doctors are available but in whom patients can choose to see, a dimension that workforce policies, focused on the number of physicians, largely overlook.

This matters because many patients, particularly women, prefer a doctor of their own gender for some kinds of care ([Kerssens et al., 1997](#); [Godager, 2012](#); [Walker et al., 2024](#)). Such preferences are likely to matter most where care involves intimate examinations or the disclosure of sensitive concerns, as in reproductive and gynaecological health, an area in which women often report feeling unheard or dismissed ([Department of Health and Social Care, 2022](#); [Victorian Department of Health, 2025](#)). When no female provider is available, women face a choice between travelling long distances or relying on locally available male providers. Whether this constraint shapes which health needs get addressed remains an open question.

I study this question using the Rural Women’s GP Service (RWGPS), an outreach program run by the Royal Flying Doctor Service (RFDS) in Victoria, Australia, that periodically sends a visiting female general practitioner (GP) to rural towns where all resident doctors are male. On visit days, patients can choose between the visiting female GP and the town’s local male providers; on all other days, only local male providers are available. Because gender choice is otherwise absent in these towns, the program creates a sharp change in access rather than a marginal shift in an already mixed workforce. I combine new data on the program’s clinic calendars with Australia’s Person-Level Integrated Data Asset (PLIDA), which links every publicly subsidised consultation, prescription, and test to the same individual over time. These records allow me to follow each woman across her encounters with female and male providers, on visit days and on all other days.

Two research designs answer two different questions. The first asks how women use the choice once it is available. A within-patient design compares each woman’s encounters with the visiting female GP to her own encounters with local male providers. The resulting

estimate captures the realized difference in care when gender choice is exercised, combining which needs a woman brings to each provider with how each provider responds. The second asks what changes when the choice arrives. A difference-in-differences design exploits variation in program entry across two towns.

When women can choose, gender-specific care is strongly concentrated in their encounters with the female GP. Among prescription days, genitourinary or hormone medications are between three and four times as common when the same woman sees the female GP as on her prescription days with local male providers, where the rate is around 6 percent. Among service days, the days with at least one billed medical service, cervical screening is more than ten times as common when she consults the female GP, against a rate of less than 1 percent. I refer to these as gender-specific outcomes: care where the underlying concern is tied to the patient's sex. The concentration matters for the towns as a whole: visit days account for about 8 percent of women's service days, yet roughly a quarter of all cervical screening among eligible women in these towns takes place on those days.

When the choice arrives, prescribing changes for the women who eventually use the service rather than for the town as a whole. Averaged over all women, the composition of prescribing does not change after program entry. Among women who eventually use the service, the probability of receiving a gender-specific prescription in a given month rises after entry, while the probability of consulting a GP does not change.

Several checks support reading these differences as specific to gender-specific concerns rather than as a general feature of the visiting GP. Cardiovascular and mental health prescribing, two domains that are relevant to women's health but not specific to it, show no comparable difference between prescription days with the female GP and with local male providers.¹ Routine pathology tests such as blood glucose and full blood counts show no comparable difference either, which weighs against the female GP simply being a more thorough provider. A further concern is that women behave differently with any provider from outside town, for instance because a visiting provider offers more privacy in a small community. Encounters with other non-local male providers allow me to investigate this: they share the visiting GP's outsider status but not her gender, and they show no comparable

¹The Royal Australian College of General Practitioners (RACGP) curriculum treats both as distinct contextual units that are directly linked to its women's health unit, reflecting their importance in the female burden of disease ([Royal Australian College of General Practitioners, 2022](#)). Unlike gynaecological and hormonal care, however, these domains are not inherently specific to women.

pattern.

The evidence is consistent both with women raising different concerns with the female GP and with differences in practice style. Most surveyed RWGPS patients report being comfortable discussing health concerns with the female GP. Encounters with the female GP also differ in the delivery of care: consultations are substantially longer, and the share of prescription days that include an antibiotic is roughly half that on prescription days with local male providers. Consultations with the female GP displace consultations with local male providers in the same month, but there is little evidence of displacement in subsequent months.

This paper makes three contributions. First, I contribute to the literature on patient-physician gender concordance. Much of the causal evidence comes from settings in which patients are assigned to physicians and in which discomfort with a physician of the other gender is unlikely to be the central concern, such as heart-attack survival ([Greenwood et al., 2018](#)), medical evaluations for disability benefits ([Cabral and Dillender, 2024](#)), and surgical outcomes ([Wallis et al., 2017](#)); see also [Si et al. \(2025\)](#).² In primary care, [Pruckner et al. \(2025\)](#) find little change in utilisation following quasi-random assignment to a female rather than male physician. Gender concordance may matter most where care is intimate and patients can choose whether, and for which needs, to consult a female physician. My setting captures this margin, which assignment-based designs deliberately abstract from.

Second, I contribute to the literature on physician gender and healthcare provision. Female and male physicians differ in consultation patterns, communication, prescribing, and the conditions they treat ([Roter and Hall, 2004](#); [Dahrouge et al., 2016](#); [Méndez et al., 2021](#); [Harrison et al., 2011](#)), but some of these differences shrink once patient sorting is removed ([Pruckner et al., 2025](#)). When patients can choose, the care delivered in encounters with female and male physicians differs sharply in gender-specific domains, suggesting that patient choice may be an important part of observed physician-gender differences in primary

²A related literature studies racial and socioeconomic concordance. Racial concordance increases take-up of preventive care among Black men ([Alsan et al., 2019](#)), shifts clinical decisions and outcomes in emergency and inpatient settings ([Ye and Yi, 2023](#); [Hill et al., 2023](#)), and is associated with lower newborn mortality ([Greenwood et al., 2020](#)); physician-patient matching along socioeconomic background affects the intensity and composition of care and health outcomes ([Kristiansen and Sheng, 2026](#)). [Frakes and Gruber \(2026\)](#) point out that encounter-level concordance designs may miss broader effects of workforce diversity that operate through patient choice and other community-level responses.

care.

Third, I contribute to the literature on rural health workforce policy. Expanding access to primary care in underserved communities improves health outcomes (Bailey and Goodman-Bacon, 2015; Okeke, 2023), and physician exits reduce utilisation and continuity of care (Simonsen et al., 2021; Zocher, 2024), yet policy in Australia has focused mainly on the aggregate supply of doctors (Swami and Scott, 2021). Evidence on periodic outreach is scarce and centres on specialist visits to remote communities (Gruen et al., 2006), while Lamiraud et al. (2026) show that preferences for gender-concordant GPs imply different predicted effects of adding male and female GPs. I study an actual change in the gender composition of local access, and the results suggest that physician supply and workforce gender composition may represent distinct policy margins.

The paper proceeds as follows. Section 2 describes the institutional setting. Section 3 presents the data. Section 4 presents the within-patient analysis. Section 5 presents the difference-in-differences analysis. Section 6 discusses the interpretation of the results and the displacement of care. Section 7 concludes.

2 Background: The Healthcare System in Australia

General practitioners play a central role in Australia’s healthcare system, serving as primary providers of medical care and acting as gatekeepers to specialist care. The majority of GPs operate in group practice settings (non-corporate), which are typically privately owned. Patients in Australia can freely choose their GP and visit multiple practices since they are not formally registered with a single practice.

The Australian national health scheme, *Medicare*, subsidises a portion of the expenses for services rendered by GPs, with reimbursements varying based on the length and complexity of consultations. GPs can adopt bulk billing, directly bill Medicare, or set higher charges, with patients paying any difference beyond what Medicare covers. Prescription medicines listed on the Pharmaceutical Benefits Scheme (PBS) are subsidised by the Commonwealth; medicines not listed on the PBS, or dispensed privately, are paid in full by the patient.

Although primary care access is universal in principle, the geographic distribution of GPs is deeply uneven. GP density in very remote areas is roughly half that in major cities, and attendance rates in remote communities are substantially below the national average

([Australian Institute of Health and Welfare, 2024](#)). Rural and remote communities therefore face persistent shortages of primary care physicians.

2.1 The Rural Women’s GP Service

The Rural Women’s GP Service (RWGPS) is operated by the Royal Flying Doctor Service (RFDS) Victoria. The program sends visiting female GPs to rural communities to provide general primary care, and its clinics are open to patients of all genders. RFDS characterises the service as providing both gender choice and access to a consistent visiting GP. The service is not restricted to women’s health; gender choice, rather than a specialised clinical scope, is its defining feature. In practice, around 98 percent of program patients are women, consistent with demand for gender-concordant care being particularly salient among women in these communities; the main analysis therefore focuses on female patients.

The program existed in a more limited form before 2014 and was formalised and expanded under RFDS Victoria from 2014 onwards. Program locations are selected on the basis of local workforce shortages. To be eligible, a community must lack access to a female GP within a 50-kilometre radius, and a local practice must be willing to host the service. RFDS estimates that each consultation with the visiting GP saves a patient about 2.3 hours of travel, or about 2,340 hours per year across the service³. The barrier is therefore a few hours of travel rather than the multi-day journeys documented for remote Indigenous communities in northern Australia ([Gruen et al., 2006](#)). In the towns and years included in the analysis, local primary care is provided by resident male GPs, sometimes supplemented by male locum physicians; I refer to both as local male providers. The visiting RWGPS physician is therefore the only female GP available locally during program clinics.

The service is delivered by a pool of six to ten female GPs. Each GP is assigned to a particular service area and returns to the same location across successive clinics, providing patients with continuity rather than a different visiting physician on each occasion. Each clinic lasts one or two days and is held approximately every six to eight weeks. Clinics are hosted by existing local practices, which provide consulting rooms, administrative support, appointment scheduling, and follow-up care between visits. Patients can book appointments with the visiting GP in advance through the host practice. This design matches the features that evaluations of outreach identify as conditions for its success, namely predictable visits

³Estimate based on 2018 data.

integrated with local services ([Gruen et al., 2006](#)). All RWGPS consultations are bulk-billed through Medicare, so patients face no direct out-of-pocket cost at the point of care. Most consultations with local GPs in the program towns are also bulk-billed, so differences in out-of-pocket cost between the two types of provider are small.

The visiting doctors are general practitioners rather than specialists: they bill standard GP consultation items, and the service is not restricted to women’s health. The gender-specific care at the centre of the analysis, cervical screening and the prescription of hormone medications and menopausal hormone therapy, falls within the core curriculum of general practice ([Royal Australian College of General Practitioners, 2022](#)) and can be provided by any GP, including local male providers. Differences in the content of care are therefore unlikely to reflect services that only the visiting doctor is trained to provide, although individual visiting GPs may have more experience in women’s health than local providers.

Two institutional features are central to the empirical analysis. First, RFDS administrative records identify the exact dates on which the visiting GP is present in each town. On these visit days, patients can choose between the visiting female GP and local male providers, who remain available as usual. On other days, the locally available primary-care providers are male. The visits therefore simultaneously increase short-run physician capacity and introduce a female, non-local provider into the local healthcare market.

Second, according to RFDS administrative information, visit dates are determined in advance on the basis of physician availability, clinic capacity, and travel logistics, rather than contemporaneous fluctuations in local healthcare demand. This scheduling process generates recurring variation in the availability of a female GP within the same community. It does not imply that patients are randomly assigned to providers: because appointments can be booked in advance, women choose whether, and for which healthcare needs, to consult the visiting GP. This choice is part of what the within-patient estimates capture.

3 Data

3.1 Data Sources and Program Towns

The data comes from two main sources. The first is the Person-Level Integrated Data Asset (PLIDA), which links individual-level records across Australian administrative datasets and allows each person to be followed over time. The primary health data comes from Medicare

Benefits Schedule (MBS) claims and Pharmaceutical Benefits Scheme (PBS) records. MBS claims record all publicly subsidised medical services, including an anonymised provider identifier, the date of service, the location, and the service item. PBS records capture subsidised prescriptions, including an anonymised prescriber identifier, the medication, and the date of prescribing. I use the date of prescribing, so that each prescription is matched to the day of the consultation at which it was written rather than the day it was dispensed.

Through PLIDA, these records are supplemented with demographic information from the 2016 Australian Census, including age, sex, and highest educational attainment, and with individual income from Australian Taxation Office records for the period 2011–2024.

The second source is a new dataset from RFDS Victoria containing RWGPS clinic calendars, which record the dates on which the visiting female GP held a clinic in each town. To my knowledge, the program has not previously been studied with administrative data. RFDS additionally provides survey responses from program users that I explore in Section 6.

Because every MBS claim and PBS record carries a provider identifier and a date, encounters with the visiting female GP can be attributed by linking the RWGPS clinic calendar to the claims: the female GP appears as a recurring provider who bills in the program town only on scheduled visit days. I construct two treatment indicators. The first, FemaleGP_{itd} , equals one if patient i consults, or receives a prescription from, the visiting female GP on date d . The second, VisitDay_{td} , equals one if the visiting female GP held a clinic in town t on date d , capturing female GP availability at the town-day level. This indicator is used in the analysis of local male providers on visit days and in the town-day analysis in Appendix B.1.

As of 2025, the Rural Women’s GP Service operates in ten Victorian towns. The analysis focuses on six program towns observed between 2011 and 2024 (Figure 1): five of the ten currently active towns, plus one former program town that exited in 2018. Of the five currently active towns not included, three are excluded because they entered the program after the end of the study period. The remaining two are excluded because they joined the program on the grounds that they had no resident GP at all, rather than no female GP, so their local healthcare environment differs substantially from that of the other program towns.

Among the six included towns, two entered the program during the sample period, with first RWGPS visits in September 2016 and January 2018 respectively. For these two towns, observations in all analyses begin in October 2015 and August 2015 respectively. These are

the earliest dates at which the all-male composition of the local resident workforce can be confirmed, since the gender of local providers cannot be verified in earlier years, and both towns are therefore observed before their first RWGPS visit.⁴

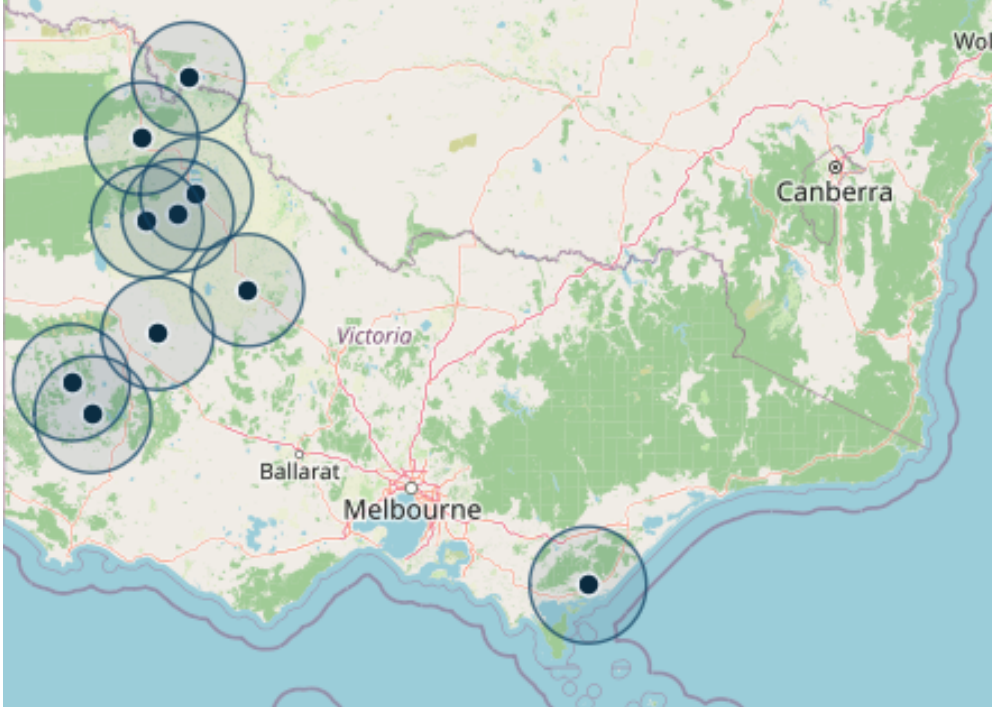


Figure 1: Map of current towns with an active program in 2025

3.2 Prescription Sample

The prescription sample is restricted to female patients resident in program towns aged between 14 and 84. Each observation is a patient-day on which at least one PBS-subsidised prescription was written; the sample comprises approximately 236,100 such patient-days. Outcomes indicate whether any prescription written that day falls in a given therapeutic category. The estimates therefore describe the composition of prescribing on prescription days, not the number of prescription days. I examine four categories: genitourinary medications (ATC chapter G), hormone medications (ATC code G03, a subset of the genitourinary category reported separately given its substantive importance), mental health medications (ATC codes N05, N06), and cardiovascular medications (ATC code C). Because hormone medications are included in both the genitourinary and hormone medication categories,

⁴One town exited the program in 2018, restricting its observations to 2011–2018; a second town has administrative data available only through 2023. Neither affects the within-patient estimates, which compare each woman to herself and rely only on the subset of town-years in which she is observed.

these two outcomes are not mutually exclusive. To examine practice style in Section 6, I also construct an indicator for antibiotics, defined as antibacterials for systemic use (ATC code J01). Appendix Table A1 lists the full correspondence between categories and ATC codes.

Within the hormone medications category, I further distinguish two age-specific sub-categories. Menopausal hormone therapy is defined as ATC codes G03C, G03D, and G03F among women aged 45 and over. Contraceptives are defined as ATC code G03A among women under 45. The contraceptive sub-category is subject to an important data limitation: most contraceptive items in Australia, including the oral contraceptive pill, were commonly dispensed privately rather than through the PBS during the sample period. Estimates for this sub-category therefore capture only PBS-subsidised contraceptive prescribing and should not be interpreted as the total difference in contraceptive prescribing. The menopausal hormone therapy sub-category does not face this limitation, since hormone replacement therapy is predominantly PBS-subsidised.

3.3 Cervical Screening Sample

The cervical screening sample is restricted to women resident in program towns aged 25 to 74, consistent with current Australian cervical screening guidelines. Each observation is a service day, a patient-day on which at least one MBS item was billed, and the outcome is an indicator equal to one if the patient received cervical screening anywhere in Australia on that day. The sample comprises approximately 374,000 service days. Prior to December 2017, guidelines additionally recommended screening from age 18, so this restriction may exclude a small number of screening events among younger women in the pre-2017 period. Cervical screening is identified using MBS items capturing both cytology-based and HPV-based screening, defined consistently across Australia’s transition to HPV-based screening in December 2017. Appendix Table A2 lists the full set of cervical screening item numbers.

3.4 Consultation Sample

From the full MBS claims data, I construct a consultation sample containing GP consultation items for both female and male patients. This sample is used to examine consultation-based outcomes and to describe the demographic characteristics of patients using primary care in the program towns. I restrict it to residents of program towns aged between 14 and

84, observed over 2011 to 2024, and construct a patient-day dataset in which each observation is a day with a GP consultation. I drop observations where a patient saw multiple providers on the same day, removing 622 observations (0.13 percent of the sample). The consultation sample comprises approximately 481,000 patient-days across the six program towns.

I identify GP consultations using standard MBS items 00003, 00023, 00036, and 00044. The consultation-length outcome, examined in Section 6, is an indicator for a long consultation, defined as a consultation exceeding 20 minutes; the threshold corresponds to the distinction between standard (Level B) and long (Level C and above) consultation items under the MBS fee schedule.

3.5 Summary Statistics

Table 1 presents summary statistics for consultation characteristics and patient demographics in the consultation sample, which covers all patients using primary care in the program towns. Because the unit of observation is the consultation day, the demographic statistics are weighted by how often patients consult. Twenty percent of consultations exceed 20 minutes. The average patient age across consultations is 56 years, and is slightly lower for women than for men. Educational attainment, recorded from the 2016 Australian Census, is concentrated in the middle of the distribution: 59 percent of consultations involve patients who have completed Year 10 to 12 schooling or hold a Certificate III or IV qualification, while 22 percent involve patients with schooling no higher than Year 9. Consultations by women are roughly twice as likely as those by men to involve a patient with a Diploma or University qualification (24 percent versus 12 percent).⁵ Table 2 presents prescription and cervical screening outcomes for female patients. Cardiovascular medications are the most common category, included on 24 percent of prescription days, followed by mental health medications (10 percent). The gender-specific categories central to the analysis, genitourinary medications and hormone medications, are each included on approximately 6 percent

⁵Because education is drawn from the 2016 Census, it is available only for patients who can be linked to a Census record and reflects attainment as of 2016. Patients who enter the sample after 2016, or who could not be matched to a Census record, or who had not yet completed their education by 2016 have missing education values. Education is therefore used only in descriptive comparisons and balance checks, not as a control variable in the main specifications, which rely on patient fixed effects to absorb all time-invariant characteristics including education.

of prescription days. Cervical screening occurs on fewer than 1 percent of service days, consistent with screening being an infrequent event even among eligible women. Additional summary statistics by ever-seen status are provided in Appendix Tables A3 and A4.

Table 1: Summary Statistics: Consultation Sample

	All			Female			Male		
	Mean	SD	N	Mean	SD	N	Mean	SD	N
<i>Panel A: Consultation characteristics (MBS)</i>									
Long consultation	0.20	0.40	481,329	0.21	0.41	273,366	0.18	0.39	207,963
<i>Panel B: Patient demographic characteristics</i>									
Age	55.78	18.60	481,329	54.29	18.97	273,366	57.75	17.90	207,963
Total income (\$AUD) ^a	44,499	51,322	275,383	40,736	46,622	155,991	49,415	56,504	119,392
Education ^b									
Year 9 or below	22.3%	—	94,130	19.7%	—	46,411	25.6%	—	47,719
Year 10–12 / Certificate	59.2%	—	249,746	56.4%	—	132,872	62.7%	—	116,874
Diploma	8.8%	—	37,011	11.2%	—	26,386	5.7%	—	10,625
University	9.7%	—	41,104	12.7%	—	29,920	6.0%	—	11,184

Notes: The unit of observation is the patient-day with a GP consultation (MBS sample), restricted to RFDS program towns.

^a Income is total annual income reported in PLIDA and is available only for working individuals aged 18 and above.

^b Education categories are derived from the Australian Census Level of Highest Educational Attainment (HEAP) classification. “Year 9 or below” corresponds to codes 811, 812. “Year 10–12 / Certificate” combines codes 611, 613, 621 with codes 721, 724 and codes 511, 514. “Diploma” combines codes 411, 413, 421. “University” combines codes 310, 211, 221, 110, 111, 114, and 120. Shares are reported in place of mean and standard deviation, computed over observations with non-missing education (421,991 in total; 235,589 female and 186,402 male).

Table 2: Summary Statistics: Prescription and Cervical Screening Outcomes (Female Patients)

	Mean	SD	N
<i>Panel A: Prescription outcomes (PBS)</i>			
Cardiovascular medications	0.24	0.43	236,106
Mental health medications	0.10	0.29	236,106
Genitourinary medications	0.06	0.24	236,106
Hormone medications	0.06	0.23	236,106
Menopausal hormone therapy (women 45+) ^a	0.02	0.13	236,106
Contraceptives (women <45) ^a	0.03	0.18	236,106
<i>Panel B: Cervical screening</i>			
Cervical screening (Pap or HPV test)	0.008	0.087	373,739

Notes: Sample restricted to female patients in RFDS program towns. Panel A reports the share of prescription days (patient-days with at least one PBS-subsidised prescription) that include a prescription in each category. Panel B reports the share of service days (patient-days with at least one MBS item) that include cervical screening, for women aged 25–74.

^a Sub-categories of hormone medications (ATC code G03). Menopausal hormone therapy comprises ATC codes G03C, G03D, and G03F among women aged 45 and over. Contraceptives comprise ATC code G03A among women under 45; because most contraceptive items in Australia are dispensed privately, this figure understates true contraceptive prescribing. Means for the two sub-categories are computed over all female prescription days.

4 Within-Patient Analysis

This section asks an encounter-level question: how does the content of the same woman’s care differ when she consults the visiting female GP rather than a local male provider? The design compares each woman to herself across encounters with different providers, holding fixed her time-invariant characteristics.

4.1 Empirical Strategy

I estimate the following equation:

$$y_{itd} = \alpha + \beta \text{FemaleGP}_{itd} + \mu_i + \lambda_{td} + \varepsilon_{itd}, \quad (1)$$

where y_{itd} denotes an outcome for patient i in town t on date d , FemaleGP_{itd} equals one if the patient consulted the visiting female GP, μ_i denotes patient fixed effects, and λ_{td} denotes town-by-date fixed effects. Standard errors are clustered at the patient level.

Patient fixed effects absorb each woman’s time-invariant characteristics. Comparisons across women are therefore not the object of this design: ever-seen and never-seen women differ systematically in observable characteristics (Table A5), and the within-patient design instead compares each woman only to herself.

The town-by-date fixed effects address a separate confound. On visit days, both physician supply and physician gender change simultaneously, since the visiting female GP adds capacity to a town that previously had only male providers. A comparison of visit days to non-visit days would therefore conflate a gender-specific response with a general supply response. With town-by-date fixed effects, β is identified from differences, on the same day in the same town, between encounters with the female GP and encounters with local male providers, so physician supply is held fixed within each comparison. As discussed in Section 2, visit dates are scheduled in advance based on physician availability and travel logistics rather than short-run local healthcare demand.

The parameter of interest, β , is the realized difference in the content of care when a woman consults the visiting female GP rather than a local male provider. It is identified from variation across the days on which the same woman is observed, net of factors common to all patients seen in her town on each date. The design does not assume that women select providers randomly: β combines the needs a woman chooses to bring to each provider with differences in the care provided during the encounter. The two channels are not separately identified.

Because the samples contain only prescription days or service days, β measures a difference in the composition of care on those days. A higher share of prescription days with the female GP that include a hormone medication can reflect more hormone prescribing, fewer prescriptions in other categories, or both. The same holds for cervical screening, which is measured on service days. The estimates therefore describe what encounters with the female

GP contain, not how much care women receive in total.

4.2 Specification Checks

Two alternative explanations would undermine the interpretation of the within-patient estimates as reflecting gender-specific care. First, the female GP could simply be a more thorough provider in general, ordering more of everything regardless of the presenting concern. Second, visit days could function as a de facto women’s-health day, with care changing for every patient in the clinic that day regardless of which provider they see.

Two results address these concerns. Against the first, routine pathology ordering shows no difference when a woman consults the female GP rather than a local male provider (Appendix Table A7). Against the second, I restrict the sample to encounters with local male providers and compare visit days with other days. Consultation length and cervical screening do not change, and the prescription estimates, at 0.003, are marginally significant but amount to roughly two percent of the corresponding within-patient differences (Table A11). Changes in the behaviour of local male providers on visit days therefore do not explain the main results. Corroborating evidence at the town-day level is reported in Appendix B.1.

A further check is reported with the discussion in Section 6: a placebo analysis in which treatment is assigned to a randomly drawn non-local male provider outside the program.

4.3 Results

4.3.1 Prescription Outcomes: Genitourinary and Hormone Medications

Table 3 reports within-patient prescription results, moving from the broadest genitourinary category through progressively narrower sub-categories. A prescription day with the female GP is 16 percentage points more likely to include a genitourinary medication than the same woman’s prescription days with local male providers (Column 1), where the rate is 6 percent: about three and a half times as likely. This difference is almost entirely accounted for by hormone medications. The coefficient on hormone medications is 15 percentage points (Column 2), against 5 percent on prescription days with local male providers, or almost four times as likely, leaving little room for non-hormonal genitourinary medications to contribute.

Columns 3 and 4 decompose the hormone medication difference into its two age-specific sub-categories. Among women aged 45 and over, a prescription day with the female GP

is 13 percentage points more likely to include menopausal hormone therapy (Column 3), against a baseline of 3 percent on prescription days with local male providers, or more than five times as likely. Among women under 45, the corresponding difference for contraceptives is 18 percentage points, against a baseline of 11 percent (Column 4). Because many contraceptive items in Australia are dispensed outside the PBS, this estimate captures only PBS-subsidised contraceptive prescribing and should not be interpreted as the total difference in contraceptive prescribing. Both sub-categories contribute to the hormone medication result, with menopausal hormone therapy showing the larger proportional difference.

Cardiovascular and mental health prescribing show no comparable difference between prescription days with the female GP and with local male providers. The coefficients are small and statistically insignificant, 0.011 and 0.013 against baselines of 25 and 8 percent respectively, an order of magnitude below the gender-specific differences (Appendix Table A6). I examine these as comparison domains because both fall within the broader women’s health framework set out in the RACGP curriculum ([Royal Australian College of General Practitioners, 2022](#)), and gender-concordance effects have been documented even in cardiac care ([Greenwood et al., 2018](#)). Unlike gynaecological and hormonal care, however, the underlying health concern is not itself gender-specific. If encounters with the female GP differed because of a general shift in women’s healthcare engagement toward the visiting provider, similar differences might be expected in these domains. The null results, alongside the large differences in genitourinary and hormone medications, support the interpretation that the concentration of care in encounters with the female GP is confined to gender-specific domains.

Table 3: Within-Patient Prescription Outcomes: Genitourinary and Hormone Medications (Women Only)

	(1)	(2)	(3)	(4)
	Genitourinary medications (All G codes)	Hormone medications (G03)	Menopausal hormone therapy (G03C/D/F, women ≥ 45)	Contraceptives (G03A, women < 45)
Female GP	0.156*** (0.011)	0.152*** (0.011)	0.126*** (0.011)	0.184*** (0.029)
Observations	236,106	236,106	179,219	56,887
R-squared	0.37	0.38	0.30	0.57
Baseline mean (ever seen)	0.061	0.054	0.029	0.112

Notes: Sample restricted to female patients. The unit of observation is the prescription day, a patient-day with at least one PBS-subsidised prescription. The outcome is an indicator for any prescription written that day falling in the specified category. Column 1 reports the full genitourinary category (ATC chapter G). Column 2 reports the hormone medications sub-category (ATC code G03), which is a subset of Column 1. Column 3 restricts Column 2 to women aged 45 and over and to ATC codes G03C, G03D, and G03F (menopausal hormone therapy). Column 4 restricts to women under 45 and ATC code G03A (contraceptives). All columns report within-patient estimates with patient and town-by-date fixed effects; standard errors clustered at patient level. Baseline means are calculated on prescription days with local male providers among ever-seen women. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.3.2 Cervical Screening

Table 4 reports the cervical screening result, the largest proportional difference in the paper. A service day with the female GP is 12 percentage points more likely to include cervical screening than the same woman’s service days with local male providers, where the rate is 1 percent across all women and 0.5 percent among ever-seen women. Even against the higher of the two baselines, screening is more than ten times as likely.

The concentration is also visible in the raw data. Visit days account for about 8 percent of eligible women’s service days in the program towns, but the screening rate on those

days is 2.5 percent, against 0.6 percent on other days (Appendix Table [A4](#), Panel B). Roughly a quarter of all cervical screening among these women over the sample period therefore takes place on visit days.⁶ The pattern parallels evidence from specialist outreach in remote Australia, where visiting clinics drew patients who presented on clinic days with problems they had not previously raised, and where patients referred to outreach clinics were disproportionately women with gynaecological problems ([Gruen et al., 2006](#)).

Screening rates in encounters with local male providers do not change on visit days (Table [A11](#), Panel C), indicating that the difference is concentrated in women’s encounters with the female GP rather than reflecting a broader visit-day change in screening. This margin is particularly relevant for cancer prevention. Research linking screening, cancer, and death records shows that 72 percent of cervical cancers diagnosed in Australia between 2002 and 2012 occurred in women who had never screened or were lapsed screeners, and that cancers diagnosed in recently screened women were less likely to be fatal ([Australian Institute of Health and Welfare, 2025](#)). In communities where female physicians are otherwise unavailable, the female GP gives women the option of being screened by a provider of the same gender.

⁶Computed from the rounded means and counts in Appendix Table [A4](#); allowing for rounding, the share lies between 25 and 29 percent. Visit days include encounters with local male providers, and the calculation covers all years in the sample, including years before program entry in the two towns that entered during the sample period.

Table 4: Cervical Screening: Within-Patient Estimates

	(1)
	Cervical screening
Female GP	0.123*** (0.006)
Observations	371,465
R-squared	0.11
Baseline mean	0.010
Baseline mean (ever seen)	0.005

Notes: Sample restricted to women aged 25–74. The unit of observation is the service day, a patient-day with at least one MBS item. The outcome is an indicator equal to one if the patient received cervical screening anywhere in Australia on that day. Within-patient estimates with patient and town-by-date fixed effects; standard errors clustered at patient level. Baseline means are calculated on service days with local male providers, for all women and for ever-seen women respectively. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5 Difference-in-Differences

The within-patient design shows how the content of care differs when women see the female GP, but every woman contributing to that comparison has used the program at least once. This section provides complementary evidence at the access margin, comparing changes in prescribing and screening before and after program entry in one treated town with changes in a not-yet-treated control town.

5.1 Empirical Strategy

Setting and samples. I exploit program entry across two towns that entered during the sample period: Town A in September 2016 and Town B in January 2018. I restrict the sample to October 2015 through December 2017, so that Town A is observed before and after entry while Town B serves as a not-yet-treated control.⁷ Because only one town enters the program within this window, treatment timing is homogeneous within the estimation sample, and concerns about heterogeneous treatment effects under staggered adoption (Callaway and Sant’Anna, 2021) do not apply. The unit of observation is the patient-month for prescription outcomes and the patient-quarter for cervical screening.⁸ The main samples include only patient-months with at least one PBS-subsidised prescription and patient-quarters with at least one billed service, so that, as in the within-patient design, the estimates describe the composition of care. To examine whether care also changes in volume, I additionally use patient-month panels that include months without care. The consultation panel covers every month of the estimation window. The prescription panel covers, for each woman, the months between her first and last observed prescription in the window, and months in between without a prescription are coded as zero; earlier and later months are excluded because residence in the town cannot be confirmed for them.

Specification. The baseline specification pools all women:

$$y_{it} = \alpha + \beta \text{DiD}_{it} + \gamma \text{Post}_t + \mu_i + \theta_{m(t)} + \delta_{y(t)} + \varepsilon_{it}, \quad (2)$$

where y_{it} denotes an outcome for patient i in period t , $\text{DiD}_{it} \equiv \text{Treated}_i \times \text{Post}_t$, Treated_i equals one for residents of Town A, and Post_t equals one from September 2016 onward. Patient fixed effects μ_i absorb the main effect of Treated_i . For prescription outcomes, $\theta_{m(t)}$ and $\delta_{y(t)}$ denote month and year fixed effects; for screening, they denote quarter and year fixed effects. The coefficient β measures the post-entry change in Town A relative to Town B across all women.

Because the female GP visits only periodically and is used by a minority of residents, a change among users may be diluted in the pooled estimate. I therefore also estimate

⁷The analysis starts in October 2015, the earliest date at which the all-male composition of the local resident workforce can be confirmed in both towns.

⁸Cervical screening is a relatively infrequent, point-in-time outcome; I therefore aggregate screening outcomes at the quarterly level.

equation (2) separately for ever-seen and never-seen women. EverSeen_i equals one if patient i consults the visiting female GP at any point between 2011 and 2024; it is constructed before restricting the sample to the estimation window, so in Town B it identifies women who use the service after that town’s entry in 2018. The estimate for ever-seen women therefore compares ever-seen women in Town A with ever-seen women in Town B. Because EverSeen_i is defined by later program use, the comparison between the two groups is descriptive heterogeneity rather than a causal contrast between ex ante user types. Appendix Table A16 reports an alternative that interacts DiD_{it} with EverSeen_i in a single regression.

Identification. Identification of the post-entry change requires that, absent the program, outcomes in the treated town would have evolved in parallel with those in the control town. I provide three pieces of evidence on this assumption. First, pre-entry patient characteristics are comparable across towns in gender, income, and education, with a modest age difference of around two years (Appendix Tables A13–A15).

Second, I estimate an event-study specification in which the treated-town indicator is interacted with event-time indicators. For prescription outcomes, I aggregate the data to two-month periods to match the program’s six-to-eight-week visit cycle and omit the two-month period immediately preceding program entry as the reference period:

$$y_{it} = \mu_i + \sum_{\substack{k \in \mathcal{W} \\ k \neq -1}} \alpha_k \mathbf{1}\{\text{EventTime}_{it} = k\} + \sum_{\substack{k \in \mathcal{W} \\ k \neq -1}} \gamma_k [\text{Treated}_i \times \mathbf{1}\{\text{EventTime}_{it} = k\}] + \varepsilon_{it}, \quad (3)$$

where k indexes two-month periods relative to program entry and $\mathcal{W}$ is the set of such periods in the estimation window. Event time is measured relative to Town A’s entry in both towns, so the event-time indicators absorb all calendar variation common to the two towns, and month and year fixed effects are not included separately. Pre-entry coefficients γ_k that are small and statistically insignificant are consistent with parallel trends. Figure 2 plots the estimates for ever-seen women, comparing ever-seen women in the two towns, and shows no differential pre-trends within this subgroup. Appendix Figure A3 reports the corresponding event studies for all women: coefficients fluctuate around a similar level before and after entry, with no change at program entry; because the reference period lies slightly above this level, most coefficients are below zero.⁹

⁹The cervical screening event study, for all women, is reported in Appendix Figure A1, with the quarter before entry as the reference period. An event study for contraceptives is not reported: the estimation

Third, I assign a placebo entry date of April 2016 and restrict the sample to the pre-entry period through August 2016 (Appendix Table A20). The placebo coefficients are statistically insignificant across all outcomes.

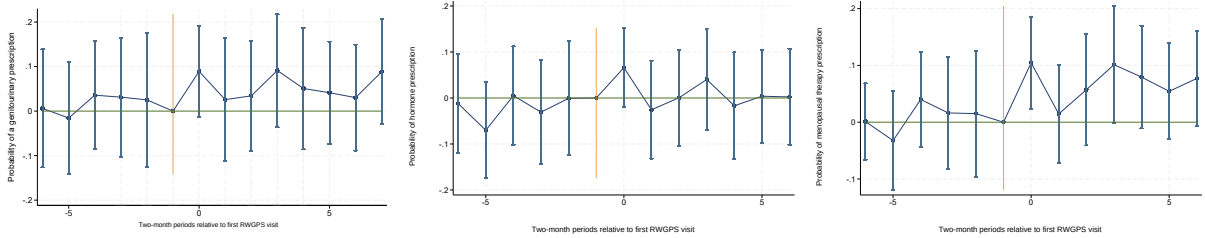


Figure 2: Event-study estimates for genitourinary medications (left), hormone medications (centre), and menopausal hormone therapy (right) among ever-seen women in the treated and control towns. The horizontal axis measures two-month periods relative to the first RWGPS visit; the reference period is the two-month period before entry.

5.2 Results

Composition of prescribing. Table 5 reports the estimates. Pooled across all women, the share of prescription months that include a genitourinary medication, a hormone medication, or menopausal hormone therapy does not change after program entry (Panel A). The change is concentrated among ever-seen women. Among them, the share of prescription months that include a genitourinary medication rises by 3.6 percentage points after entry, against a pre-period mean of 8 percent, and the share that include menopausal hormone therapy rises by 4.6 percentage points, against a pre-period mean of 3 percent (Panel B).¹⁰ Estimates for never-seen women are close to zero (Panel C). Appendix Table A16 reports the interacted specification, which constrains ever-seen and never-seen women in the control town to share the same post-period change. The interaction terms are positive and significant at the 5 percent level for genitourinary medications, hormone medications, and menopausal hormone therapy, but the implied changes for ever-seen women are smaller than in Panel B and significant only for menopausal hormone therapy, at the 10 percent sample is small (3,489 patient-months) and the PBS captures only a minority of contraceptive dispensing. The event study for consultation length, discussed in Section 6, is reported in Appendix Figure A2.

¹⁰The estimate for hormone medications is similar in size (3.3 percentage points) but not statistically significant. The contraceptive estimate is negative and significant at the 5 percent level, but it rests on 352 patient-months, captures only PBS-subsidised items, and exceeds the pre-period mean in absolute value (0.138 against 0.106), which would imply a negative post-period rate; I therefore do not interpret it.

level. Appendix Table [A17](#) shows no differential change in mental health or cardiovascular prescribing.

Volume of prescribing and GP contact. The increase among ever-seen women is not only a change in shares. Once months without a prescription between a woman’s first and last observed prescription are counted as zero, the monthly probability that an ever-seen woman receives a genitourinary medication rises by 2.2 percentage points, against a pre-period mean of 3.1 percent, and the probability that she receives menopausal hormone therapy more than doubles, from 1.2 percent (Panel D). Over the same period, these women are no more likely to consult a GP in a given month, although they are 7.8 percentage points more likely to receive any PBS-subsidised prescription, against a pre-period mean of 37 percent (Appendix Table [A18](#)). Program entry is therefore associated with more gender-specific prescribing among ever-seen women without more frequent GP contact. Mental health and cardiovascular prescribing also rise, by about 45 and 13 percent of their pre-period means, but neither change is statistically significant (Panel E). Together with the rise in the probability of receiving any PBS-subsidised prescription, this suggests that part of the change reflects more prescribing in general, with the largest proportional increases in genitourinary medications and menopausal hormone therapy.

Cervical screening. For cervical screening, the difference-in-differences design is less informative than for prescribing, for two reasons. First, screening is infrequent by design: women were advised to screen every two years before December 2017 and every five years after, so within the 27-month estimation window most women are due for at most one test. Program entry can therefore change when and where a test takes place more than how many tests take place, and a quarterly estimate averages over many quarters in which no test is due. Second, the comparison rests on two towns over a short window, whereas the within-patient estimates draw on six towns over the full sample period and ask directly whether screening is included in the same woman’s encounters with the female GP. For these reasons, the within-patient design remains the primary evidence for this outcome. In the pooled specification, the probability that a quarter with a billed service includes cervical screening rises by 2.4 percentage points after entry in the treated town relative to the control town, against a pre-period mean of 2.9 percent (Appendix Table [A19](#)), and the event study places the increase mainly in the first two quarters after entry (Appendix Figure [A1](#)). In

the interacted specification, the post-entry change is 2.7 percentage points for never-seen women and 1.0 percentage point for ever-seen women, with a joint p -value of 0.447. Routine pathology does not show a parallel increase: blood glucose testing does not change, and program entry is associated with lower full blood count testing in the pooled specification (-0.039 , standard error 0.017). The ever-seen subsample is too small to distinguish a change in screening from zero, so I do not interpret the split by ever-seen status.

Table 5: Difference-in-Differences: Prescription Outcomes by Sample (Women Only)

	(1) Genitourinary medications	(2) Hormone medications	(3) Menopausal hormone therapy	(4) Contraceptives
<i>Panel A: All women, prescription months</i>				
DiD	-0.002 (0.010)	-0.003 (0.009)	0.003 (0.008)	-0.037 (0.032)
Observations	15,977	15,977	12,463	3,489
Pre-period mean (treated town)	0.089	0.082	0.031	0.198
<i>Panel B: Ever-seen women, prescription months</i>				
DiD	0.036* (0.021)	0.033 (0.020)	0.046*** (0.017)	-0.138** (0.054)
Observations	2,280	2,280	1,925	352
Pre-period mean (treated town)	0.083	0.073	0.031	0.106
<i>Panel C: Never-seen women, prescription months</i>				
DiD	-0.011 (0.011)	-0.011 (0.010)	-0.009 (0.009)	-0.010 (0.037)
Observations	13,697	13,697	10,538	3,137
Pre-period mean (treated town)	0.089	0.083	0.031	0.205
<i>Panel D: Ever-seen women, months between first and last prescription (months without a prescription coded as zero)</i>				
DiD	0.022** (0.009)	0.016* (0.008)	0.018*** (0.007)	
Observations	6,378	6,378	4,914	
Pre-period mean (treated town)	0.031	0.027	0.012	
<i>Panel E: Comparison domains, ever-seen women, months between first and last prescription</i>				
	Mental health medications	Cardiovascular medications		
DiD	0.018 (0.011)	0.015 (0.012)		
Observations	6,378	6,378		
Pre-period mean (treated town)	0.040	0.120		

Notes: Sample restricted to female patients, October 2015 through December 2017. One town enters the program in September 2016; the second town serves as a not-yet-treated control. Panels A to C use patient-months with at least one PBS-subsidised prescription; Panels D and E use patient-month panels for ever-seen women covering the months between each woman's first and last observed prescription in the window, in which months without a prescription are coded as zero. Panels B and C estimate the specification separately by ever-seen status (having seen the female GP at any point between 2011 and 2024). All specifications include the post-period main effect, patient fixed effects, month fixed effects, and year fixed effects. Standard errors clustered at patient level; with one treated and one control town, these do not account for town-level shocks. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6 Discussion

6.1 Interpreting the Differences

What explains the concentration of gender-specific care in women’s encounters with the female GP? Three explanations are plausible: women may be more comfortable raising sensitive concerns with a gender-concordant provider, the female GP may adopt a different practice style, and women may value consulting a provider who does not live in their community. The qualitative evidence, the practice-style outcomes, and the placebo-provider analysis are informative about each in turn.

Illustrative quotes from patients and providers collected in annual reports and surveys are consistent with the first explanation.

“Women come to me for women’s health issues like pap tests or menopause management, but as we build a rapport they come to me for regular care.”

“A female GP who understands post-menopausal women. You could take a list of items you wanted to discuss and she worked through each of them. Extremely thorough without feeling any sense of rushing. I felt listened to and could ask anything.”

“I was encouraged by the female GP to have a test that I had been avoiding for years.”

“It’s been really good for young girls who want somebody that they feel comfortable with, and the mums feel comfortable too.”

“Be able to talk to a female doctor. This is something we are lacking in the country.”

A common theme across these responses is the importance of comfort, trust, and communication. Patients describe feeling understood, listened to, and more willing to discuss concerns with a female physician. Survey responses point the same way: 84 percent of surveyed women gave a score of four or five on a five-point scale when asked about their comfort discussing health concerns with the female GP. These are self-reported measures from a small and potentially selected sample of program users.

A second possibility is that encounters with the female GP differ in practice style. Female and male physicians differ systematically in consultation patterns and communication, with female physicians tending to spend more time with patients and engage in more patient-centred communication ([Roter and Hall, 2004](#); [Dahrouge et al., 2016](#); [Méndez et al., 2021](#)); recent causal evidence also shows that female and male primary care physicians differ in

how they organise care when assignment is exogenous (Pruckner et al., 2025). I examine consultation length and antibiotic prescribing as two observable dimensions of how care is delivered.

Encounters with the female GP are substantially more intensive. A consultation day with the female GP is 55 percentage points more likely to involve a long consultation than the same woman’s consultation days with local male providers, against a baseline rate of 20 percent (Table 6). In the difference-in-differences design, program entry is associated with an increase in the probability of long consultations: in the interacted specification, the combined change among ever-seen women is 11 percentage points, and never-seen women show an increase of 4 percentage points (Table 7, Column 2; Appendix Figure A2 shows no differential pre-trends among ever-seen women). The pattern holds in a patient-month panel in which months without a consultation are coded as zero: the monthly probability of a long consultation rises by 5.5 percentage points among ever-seen women, against a pre-period mean of 9.2 percent, with a placebo estimate close to zero (Appendix Table A18, Panel B). Part of the within-patient difference likely reflects women bringing a more extensive set of concerns to the female GP rather than physician style alone: a consultation booked for cervical screening or menopausal care may naturally run longer. Both readings are consistent with the qualitative evidence above, where patients emphasise not being rushed and describe bringing a list of items to work through.

Antibiotic prescribing provides a second indication that encounters with the female GP differ beyond the gender-specific outcomes. A prescription day with the female GP is 10 percentage points less likely to include an antibiotic than the same woman’s prescription days with local male providers, against a baseline of 20 percent, roughly halving the rate (Table 6, Column 2). Without information on clinical indication, I do not interpret lower prescribing as evidence of better care. This result also bears on the composition interpretation in Section 4: prescription days with the female GP differ in their non-gender-specific content, so part of the higher share of gender-specific items may reflect differences in other prescribing rather than only additional gender-specific prescriptions. In the difference-in-differences design, antibiotic prescribing shows no significant change for either ever-seen or never-seen women following program entry,¹¹ consistent with a provider-specific prescribing style rather than a town-wide change in prescribing practice.

¹¹The antibiotic DiD coefficient is 0.018 (SE 0.017) and the ever-seen interaction is 0.001 (SE 0.024).

The third explanation relates to privacy and confidentiality in small communities: consulting a visiting physician allows women to discuss sensitive concerns with someone who does not live in town. The placebo-provider analysis speaks to this. Assigning treatment to a randomly drawn non-local male provider outside the program yields estimates that are near zero for all outcomes except consultation length, where the estimate is small and negative (Tables A8, A9, and A10). The main patterns are therefore not replicated for other non-local physicians, suggesting that non-local provider status alone is unlikely to explain them. Because these providers are not part of the program, the test does not hold fixed other features of the RWGPS, such as its scheduling and its presentation as a women’s service.

Taken together, the evidence is consistent with both patient-side and provider-side channels. Women’s accounts point to comfort raising sensitive concerns with the female GP, while consultation length and antibiotic prescribing indicate that encounters with the female GP are also delivered differently.

6.2 Displacement of Care

I further examine whether consultations with the female GP displace subsequent care from local male providers. I restrict attention to ever-seen women and examine whether their consultations with local male providers change in the months following a consultation with the female GP. If consultations with the female GP substitute for care from local male providers, consultations with local male providers should fall in subsequent periods.

The results show substitution within the month but little evidence of persistent displacement. Among ever-seen women, consulting the female GP in a given month is associated with a 65 percentage point reduction in the probability of a consultation with local male providers in that same month, but the lagged effects over the following three months are near zero and statistically insignificant (Appendix Table A12). Access to the female GP therefore changes with whom some care occurs, with little evidence that it draws women away from local male providers beyond the month of the visit.

Program entry is also associated with a higher probability of a long consultation among never-seen women and among male patients, who almost never use the program (Table 7, Columns 2 and 3). This pattern is consistent with local male providers reallocating consultation time, although the design cannot identify the channel.

Table 6: Practice Style: Within-Patient Estimates (Women Only)

	(1)	(2)
	Long consultation	Antibiotics
Female GP	0.549*** (0.009)	-0.105*** (0.009)
Observations	272,022	236,106
R-squared	0.218	0.268
Baseline mean	0.20	0.20
Baseline mean (ever seen)	0.20	0.20

Notes: Sample restricted to female patients. Column 1 uses the MBS consultation sample; the outcome is an indicator for consultations exceeding 20 minutes. Column 2 uses the PBS prescription sample; the outcome is an indicator for any antibiotic (ATC code J01) prescribed on the day, among prescription days. Within-patient estimates with patient and town-by-date fixed effects; standard errors clustered at patient level. Baseline means are calculated on days with local male providers, for all women and for ever-seen women respectively. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 7: Difference-in-Differences: Consultation Length

	(1)	(2)	(3)
	All patients	Women	Men
DiD	0.057*** (0.012)	0.042*** (0.016)	0.073*** (0.019)
DiD $\times$ Ever seen	0.055** (0.018)	0.072*** (0.020)	-0.050 (0.091)
Effect (ever seen)	0.112	0.114	0.023
Pre-period mean (treated town)	0.17		
Pre-period mean (control town)	0.28		
Observations	32,569	18,470	14,099
R-squared	0.230	0.219	0.244

Notes: Sample restricted to October 2015 through December 2017. The unit of observation is the patient-month with at least one GP consultation. One town enters the program in September 2016; the second town serves as a not-yet-treated control. Estimates from the interacted specification (see Appendix Table A16). Ever seen is an indicator for patients who see the female GP at any point between 2011 and 2024. Effect (ever seen) is the sum of the DiD coefficient and the interaction term. Specifications include the post-period main effect, patient fixed effects, month fixed effects, and year fixed effects. Standard errors clustered at patient level; with one treated and one control town, these do not account for town-level shocks. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7 Conclusion

This paper examines how the content of women’s primary care differs when they gain the choice of a female physician. Exploiting a program that periodically brings a female GP to rural towns where all resident GPs are male, I compare each woman’s encounters with the female GP to her encounters with local male providers. Gender-specific care is strongly

concentrated in women’s encounters with the female GP. Among prescription days, genitourinary or hormone medications are between three and four times as common when the same woman sees the female GP, and among service days, cervical screening is more than ten times as common. Cardiovascular and mental health prescribing and routine pathology show no comparable differences. A complementary difference-in-differences analysis finds no change in prescribing composition averaged over all women, while among women who eventually use the service, program entry is associated with increases in gender-specific prescribing even when months without a prescription are included, without an increase in the probability of GP contact.

The evidence is consistent with both patient-side and provider-side channels. Most surveyed program users report being comfortable discussing health concerns with the female GP, and encounters with her are longer and less likely to involve an antibiotic. Encounters with other non-local male providers show no comparable pattern. Care shifted to the female GP displaces consultations with local male providers only in the month of the visit.

Some limitations qualify these conclusions. The within-patient estimates combine the needs women bring to the female GP with the care she provides, and they do not separate her gender from other characteristics such as training or consultation style. They also describe the composition of care on prescription days and service days rather than its total volume; the monthly difference-in-differences estimates speak to volume, but only for ever-seen women. Finally, the difference-in-differences analysis compares a single treated town with a single control town over a short period, which rules out conventional inference at the level of treatment assignment.

Because the relevant barrier in these towns is a few hours of travel rather than extreme remoteness, the findings speak to rural communities where a female physician is within reach but not locally available, not only to the most remote areas. These findings suggest that physician supply and workforce gender composition are not equivalent policy levers. Where access to female physicians is limited, expanding physician capacity without expanding gender choice may not generate the same changes in which healthcare needs are addressed.

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A Data Construction and Additional Results

A.1 Data Construction and Additional Descriptive Statistics

Outcome Definitions

Table A1: PBS Medication Categories

Category	ATC Code	Description
Genitourinary medications	G01, G02, G03, G04	Gynaecological anti-infectives, other gynaecologicals, hormone medications, urologicals
Hormone medications	G03	Contraceptives, hormone replacement therapy (subset of genitourinary medications)
Menopausal hormone therapy	G03C, G03D, G03F	Restricted to women aged 45 and over
Contraceptives	G03A	Restricted to women under 45; predominantly dispensed privately
Mental health medications	N05, N06	Anxiolytics, antidepressants
Cardiovascular medications	C	Antihypertensives, statins
Antibiotics	J01	Antibacterials for systemic use

Table A2: MBS Pathology Items

Category	Item Numbers	Description
Cervical screening (post 2017)	73053, 73055, 73070–73076	HPV test
Cervical screening (pre 2017)	69418, 69419, 73053, 73055, 73057, 73069	Pap test
Full blood count	65070, 65071	Routine haematology
Blood glucose	66500, 66503, 66506	Routine health check

Additional Summary Statistics

Table A3: Summary Statistics by Subgroup: Consultation Outcomes

	Ever seen			Never seen			Visit days			Non-visit days		
	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD	N
Long consultation	0.24	0.42	108,701	0.19	0.39	372,628	0.29	0.45	38,321	0.19	0.39	443,008

Notes: The unit of observation is the patient-day with a GP consultation (MBS sample), restricted to RFDS program towns. Ever-seen patients are those who consult the visiting female GP at least once between 2011 and 2024. Visit days are days on which the visiting female GP held a clinic in the town.

Table A4: Summary Statistics by Subgroup: Prescription and Cervical Screening Outcomes (Female Patients)

	Ever seen			Never seen			Visit days			Non-visit days		
	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD	N
<i>Panel A: Prescription outcomes (PBS)</i>												
Mental health medications	0.08	0.27	45,284	0.10	0.30	190,822	0.09	0.28	18,781	0.10	0.29	217,325
Cardiovascular medications	0.25	0.43	45,284	0.24	0.43	190,822	0.25	0.43	18,781	0.24	0.43	217,325
Genitourinary medications	0.07	0.26	45,284	0.06	0.24	190,822	0.09	0.29	18,781	0.06	0.24	217,325
Hormone medications	0.07	0.25	45,284	0.05	0.22	190,822	0.08	0.28	18,781	0.05	0.22	217,325
Menopausal hormone therapy	0.03	0.17	45,284	0.02	0.12	190,822	0.04	0.19	18,781	0.02	0.13	217,325
Contraceptives	0.03	0.17	45,284	0.03	0.18	190,822	0.04	0.20	18,781	0.03	0.18	217,325
<i>Panel B: Cervical screening</i>												
Cervical screening (Pap or HPV test)	0.008	0.09	142,531	0.007	0.08	231,680	0.025	0.16	30,858	0.006	0.070	343,353

Notes: Sample restricted to female patients in RFDS program towns. Panel A uses prescription days (patient-days with at least one PBS-subsidised prescription); Panel B uses service days (patient-days with at least one MBS item). Ever-seen women are patients who consult the visiting female GP at least once between 2011 and 2024. Visit days are days on which the visiting female GP held a clinic in the town. Menopausal hormone therapy and contraceptives are the age-specific sub-categories of hormone medications defined in Table A1; their means are computed over all female prescription days.

Characteristics of Ever-Seen Women

Table A5: Balance: Ever-Seen vs Never-Seen Women

	Never seen	Ever seen	Diff.	SE	<i>p</i> -value	N (never)	N (ever)
Age	49.00	51.65	−2.65	0.191	0.000	30,645	15,396
Education	1.81	1.91	−0.10	0.012	0.000	26,178	13,771
Total income (\$AUD)	41,371	46,768	−5,398	587.8	0.000	19,048	10,970

Notes: Comparison of ever-seen women (patients who see the female GP at least once between 2011 and 2024) to never-seen women. Diff. reports the mean difference (never seen − ever seen).

A.2 Within-Patient Analysis: Additional Evidence and Validation

Comparison Domains

Table A6: Comparison Domains: Cardiovascular and Mental Health Prescribing (Women Only)

	(1)	(2)
	Mental health medications	Cardiovascular medications
Female GP	0.013 (0.008)	0.011 (0.011)
Observations	236,106	236,106
Baseline mean (ever seen)	0.08	0.25
R-squared	0.441	0.366

Notes: Sample restricted to female patients. The unit of observation is the prescription day, a patient-day with at least one PBS-subsidised prescription; the outcome is an indicator for any prescription that day in the specified category. Within-patient estimates with patient and town-by-date fixed effects; standard errors clustered at the patient level. Baseline means are calculated on prescription days with local male providers among ever-seen women. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A7: Specification Check: Routine Pathology Tests

	Blood glucose	Full blood count
Female GP	0.002 (0.004)	−0.0002 (0.004)
Observations	371,465	371,465
R-squared	0.101	0.120

Notes: All estimates are within-patient specifications with patient and town-by-date fixed effects; standard errors clustered at the patient level. The unit of observation is the service day, a patient-day with at least one MBS item; the outcome is an indicator for the corresponding pathology test being performed that day. These serve as a specification check since there is no theoretical reason for them to respond to provider gender. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Placebo-Provider Tests

Table A8: Placebo Providers: Consultation Length (Women Only)

	Long consultation
Placebo provider	−0.008*** (0.003)
Observations	247,828
R-squared	0.189

Notes: Within-patient estimates in which the visiting female GP is replaced by a placebo provider, randomly drawn among non-local male providers outside the program. Sample restricted to female patients and non-visit days. The outcome is an indicator for a long consultation. Specifications include patient and town-by-date fixed effects; standard errors clustered at the patient level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A9: Placebo Providers: Prescription Outcomes (Women Only)

	Genitourinary medications	Hormone medications	Menopausal hormone therapy	Contraceptives
Placebo provider	0.003 (0.005)	0.006 (0.005)	−0.003 (0.003)	0.026 (0.017)
Observations	215,902	215,902	163,714	46,840
R-squared	0.364	0.376	0.293	0.522

Notes: Within-patient estimates in which the visiting female GP is replaced by a placebo provider, randomly drawn among non-local male providers outside the program. Sample restricted to female patients and non-visit days. Specifications include patient and town-by-date fixed effects; standard errors clustered at the patient level. Menopausal hormone therapy and contraceptive columns restrict to women aged 45 and over and to women under 45 respectively. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A10: Placebo Providers: Cervical Screening

	Cervical screening
Placebo provider	0.001 (0.001)
Observations	340,805
R-squared	0.088

Notes: Within-patient estimates in which the visiting female GP is replaced by a placebo provider, randomly drawn among non-local male providers outside the program. Sample restricted to women aged 25–74 and non-visit days. Specifications include patient and town-by-date fixed effects; standard errors clustered at the patient level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Local Male Providers on Visit Days

Table A11: Local Male Providers on Visit Days (Women Only)

	(1)	(2)	(3)	(4)
	Long consultation	Genitourinary medications	Hormone medications	Cervical screening
<i>Panel A: Consultation length</i>				
Visit day	0.003 (0.003)			
Observations	266,203			
R-squared	0.13			
<i>Panel B: Prescriptions</i>				
Visit day		0.003* (0.002)	0.003* (0.002)	
Observations		232,582	232,582	
R-squared		0.3	0.3	
<i>Panel C: Cervical screening</i>				
Visit day				0.0004 (0.001)
Observations				367,565
R-squared				0.04
Baseline mean (Panel A)	0.19			
Baseline mean (Panel B)		0.059	0.053	
Baseline mean (Panel C)				0.006

Notes: Sample restricted to female patients and to encounters with local male providers. The coefficient compares encounters with local male providers on visit days with encounters on other days. Panel A: outcome is an indicator for a long consultation. Panel B: outcome is an indicator for any prescription in each category, among prescription days. Panel C: women aged 25–74; outcome is an indicator for cervical screening on a service day. All specifications include patient and town-by-year-month fixed effects; standard errors clustered at patient level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Displacement of Care

Table A12: Displacement of Care from Local Male Providers: Contemporaneous and Lagged Effects

	(1)	(2)
	Any consultation with local male providers	Number of consultations with local male providers
Female GP consultation (t)	−0.652*** (0.009)	−0.884*** (0.013)
Female GP consultation ($t - 1$)	−0.006 (0.005)	−0.005 (0.016)
Female GP consultation ($t - 2$)	−0.004 (0.006)	0.009 (0.017)
Female GP consultation ($t - 3$)	−0.008 (0.005)	−0.008 (0.017)
Observations	64,583	64,583
R-squared	0.685	0.276

Notes: The unit of observation is the patient-month. Sample restricted to ever-seen women. The outcome in Column 1 is an indicator for any consultation with local male providers in the month; in Column 2 it is the number of such consultations in the month. Female GP consultation (t) is an indicator for consulting the visiting female GP in the current month; ($t - 1$) to ($t - 3$) are its one- to three-month lags. All specifications include patient and town-by-year-month fixed effects; standard errors clustered at patient level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.3 Difference-in-Differences: Additional Evidence

Pre-Entry Characteristics

Table A13: Balance: DiD Sample, Treated vs Control Town

	Town B	Town A	Diff.	SE	<i>p</i> -value	N (Town B)	N (Town A)
Age	52.15	54.52	−2.37	0.926	0.011	561	1,662
Female patient	0.58	0.57	0.01	0.024	0.675	561	1,662
Education	2.15	2.15	−0.01	0.044	0.873	494	1,473
Total income (\$AUD)	46,381	42,152	4,228	2,753.5	0.125	394	1,030

Notes: Pre-period comparison of patients in Town B (control) and Town A (treated) in the DiD sample. Diff. reports the mean difference (Town B − Town A).

Table A14: Balance: DiD Sample, Ever-Seen Women by Town

	Town B	Town A	Diff.	SE	<i>p</i> -value	N (Town B)	N (Town A)
Age	52.65	55.60	−2.95	1.748	0.092	131	165
Education	2.20	2.30	−0.10	0.114	0.366	114	151
Total income (\$AUD)	41,439	37,525	3,914	4,211.9	0.354	100	109

Notes: Pre-period comparison of ever-seen women in Town B and Town A. Ever seen is defined over 2011–2024, so Town B ever-seen women are those who use the service after that town’s entry in 2018. Diff. reports the mean difference (Town B − Town A).

Table A15: Balance: DiD Sample, Ever Seen vs Never Seen

	Never seen	Ever seen	Diff.	SE	<i>p</i> -value	N (never)	N (ever)
<i>Panel A: All patients</i>							
Age	53.82	54.56	−0.75	1.166	0.523	1,915	308
Female patient	0.51	0.96	−0.45	0.029	0.000	1,915	308
Education	2.14	2.24	−0.10	0.055	0.070	1,690	277
Total income (\$AUD)	43,781	40,741	3,040	3,442.4	0.377	1,209	215
<i>Panel B: Women only</i>							
Long consultation	0.27	0.38	−0.11	0.029	0.000	943	324
Age	53.11	51.76	1.35	1.229	0.273	943	324
Education	2.26	2.27	−0.01	0.062	0.856	830	285
Total income (\$AUD)	35,804	40,601	−4,797	2,465.0	0.052	593	226

Notes: Pre-period comparison of ever-seen patients (those who see the female GP at least once between 2011 and 2024) and never-seen patients within the DiD sample, pooling the treated and control towns. Diff. reports the mean difference (never seen − ever seen).

Interacted Specification

Table A16: Difference-in-Differences: Prescription Outcomes, Interacted Specification (Women Only)

	(1)	(2)	(3)	(4)
	Genitourinary medications	Hormone medications	Menopausal hormone therapy	Contraceptives
DiD	−0.006 (0.010)	−0.007 (0.009)	−0.001 (0.008)	−0.036 (0.033)
DiD × Ever seen	0.030** (0.014)	0.029** (0.014)	0.026** (0.012)	−0.018 (0.027)
Effect (ever seen)	0.024	0.022	0.025	−0.054
Joint p -value (ever seen)	0.126	0.158	0.055	0.160
Pre-period mean (treated town)	0.089	0.082	0.031	0.198
Observations	15,977	15,977	12,463	3,489
R-squared	0.435	0.453	0.355	0.460

Notes: Estimates from equation (2) with the DiD term also interacted with ever seen, in a single regression. DiD gives the post-entry change for never-seen women; the sum of DiD and the interaction gives the change for ever-seen women. Unlike the separate estimates in Table 5, this specification constrains ever-seen and never-seen women in the control town to share the same post-period change. Sample restricted to female patients, October 2015 through December 2017. The unit of observation is the patient-month with at least one PBS-subsidised prescription. One town enters the program in September 2016; the second town serves as a not-yet-treated control. Ever seen is an indicator for patients who see the female GP at any point between 2011 and 2024. Effect (ever seen) is the sum of the DiD coefficient and the interaction term; the joint p -value tests the null that this sum equals zero. Column 3 restricts to women aged 45 and over and menopausal hormone therapy (ATC codes G03C, G03D, G03F); Column 4 restricts to women under 45 and contraceptives (ATC code G03A). Specifications include the post-period main effect, patient fixed effects, month fixed effects, and year fixed effects. Standard errors clustered at patient level; with one treated and one control town, these do not account for town-level shocks. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Additional Outcomes

Table A17: Difference-in-Differences: Comparison Domains, Interacted Specification (Women Only)

	(1)	(2)
	Mental health medications	Cardiovascular medications
DiD	0.010 (0.013)	0.010 (0.014)
DiD $\times$ Ever seen	0.007 (0.016)	-0.033 (0.021)
Observations	15,977	15,977
Pre-period mean (treated town)	0.121	0.310
R-squared	0.553	0.422

Notes: Sample restricted to female patients, October 2015 through December 2017. The unit of observation is the patient-month with at least one PBS-subsidised prescription. One town enters the program in September 2016; the second town serves as a not-yet-treated control. Ever seen is an indicator for patients who see the female GP at any point between 2011 and 2024. Specifications include the post-period main effect, patient fixed effects, month fixed effects, and year fixed effects. Standard errors clustered at patient level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A18: Difference-in-Differences: Monthly GP Contact and Prescribing, Including Months Without Care (Women Only)

	(1) Any GP consultation	(2) Long consultation	(3) Any PBS prescription
<i>Panel A: All women</i>			
DiD	0.015 (0.011)	0.032*** (0.008)	
Pre-period mean (treated town)	0.474	0.091	
Observations	40,085	40,085	
<i>Panel B: Ever-seen women</i>			
DiD	-0.007 (0.020)	0.055*** (0.015)	0.078*** (0.023)
Pre-period mean (treated town)	0.490	0.092	0.369
Observations	9,612	9,612	6,378
<i>Panel C: Never-seen women</i>			
DiD	0.027* (0.014)	0.028*** (0.010)	
Pre-period mean (treated town)	0.471	0.091	
Observations	30,473	30,473	

Notes: Each cell reports the difference-in-differences coefficient from equation (2), estimated on patient-month panels in which months without the outcome are coded as zero. The GP consultation panel (Columns 1 and 2) covers every month of the estimation window. The prescription panel (Column 3) is constructed separately from PBS records and covers the months between each woman's first and last observed prescription, so the two samples differ. The outcome is an indicator for any GP consultation in the month (Column 1), any long consultation in the month (Column 2), or any PBS-subsidised prescription in the month (Column 3; estimated for ever-seen women only). Sample restricted to female patients, October 2015 through December 2017; one town enters the program in September 2016 and the second serves as a not-yet-treated control. Panels B and C estimate the specification separately by ever-seen status (having seen the female GP at any point between 2011 and 2024). A placebo entry date of April 2016 yields an estimate for long consultations among ever-seen women of -0.006 (p -value 0.742) in the interacted specification. All specifications include the post-period main effect, patient fixed effects, month fixed effects, and year fixed effects. Standard errors clustered at patient level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A19: Difference-in-Differences: Cervical Screening and Routine Pathology

	(1)	(2)	(3)	(4)
	Cervical screening	Cervical screening	Blood glucose	Full blood count
DiD	0.024** (0.011)	0.027** (0.011)	−0.010 (0.010)	−0.039 (0.027)
DiD × Ever seen		−0.017 (0.011)	0.001 (0.012)	0.025 (0.033)
Effect (ever seen)		0.010	−0.009	−0.015
Joint p -value (ever seen)		0.447	0.465	0.776
Pre-period mean (treated town)	0.029	0.029		
Observations	9,909	9,909	9,909	9,909

Notes: Sample restricted to October 2015 through December 2017 and to women aged 25–74. The unit of observation is the patient-quarter with at least one billed MBS service. One town enters the program in September 2016; the second town serves as a not-yet-treated control. Column 1 reports the pooled specification; Columns 2 to 4 include the interaction with ever seen, an indicator for patients who see the female GP at any point between 2011 and 2024. Effect (ever seen) is the sum of the DiD coefficient and the interaction term; the joint p -value tests the null that this sum equals zero. In the pooled specification, the full blood count estimate is −0.039 (SE 0.017). Specifications include the post-period main effect, patient fixed effects, quarter fixed effects, and year fixed effects. Standard errors clustered at the patient level; with one treated and one control town, these do not account for town-level shocks.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Placebo Entry Date

Table A20: Placebo DiD Estimates

	(1)	(2)	(3)
	Genitourinary medications	Hormone medications	Cervical screening
<i>Panel A: All women, interacted specification</i>			
DiD placebo	0.015 (0.014)	0.012 (0.013)	0.023 (0.018)
DiD placebo $\times$ Ever seen	-0.007 (0.024)	0.005 (0.023)	0.010 (0.021)
Observations	6,066	6,066	3,057
<i>Panel B: Ever-seen women, months between first and last prescription (months without a prescription coded as zero)</i>			
DiD placebo	0.015 (0.014)	0.019 (0.014)	
Observations	2,342	2,342	

Notes: Placebo DiD estimates assigning program entry to April 2016 instead of September 2016, with the sample restricted to the pre-entry period through August 2016. Panel A, Columns 1 and 2, use patient-months with at least one PBS-subsidised prescription; Column 3 uses patient-quarters with at least one billed MBS service, for women aged 25–74. Panel B uses a patient-month panel of ever-seen women covering the months between each woman’s first and last observed prescription, in which months without a prescription are coded as zero. Ever seen is an indicator for patients who see the female GP at any point between 2011 and 2024. Specifications include the post-period main effect, patient fixed effects, month fixed effects, and year fixed effects (quarter fixed effects in place of month fixed effects in Column 3). Standard errors clustered at patient level.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Additional Event-Study Evidence

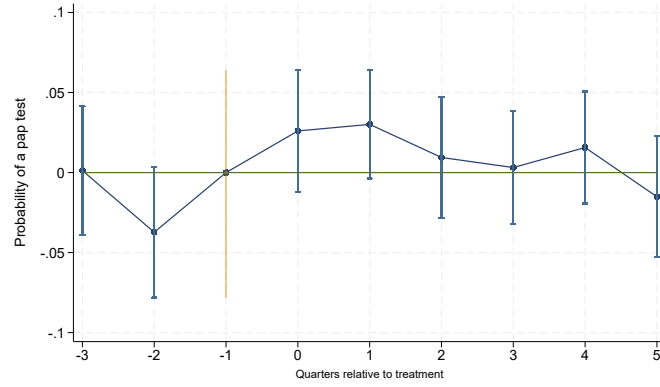


Figure A1: Event-study estimates for cervical screening among all women. The reference period is one quarter before program entry.

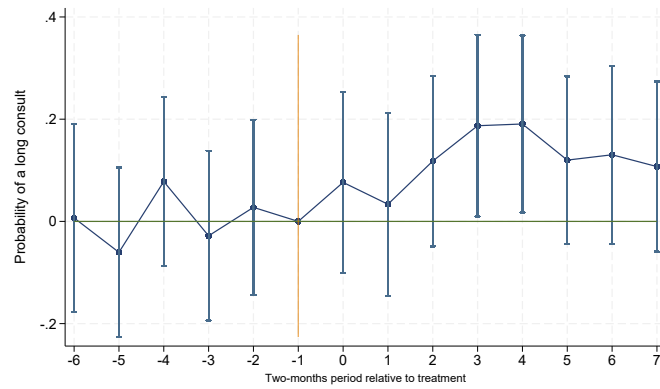


Figure A2: Event-study estimates for long consultations among ever-seen women in the treated and control towns. The reference period is two months before program entry. Consultation-length results are discussed in Section 6.

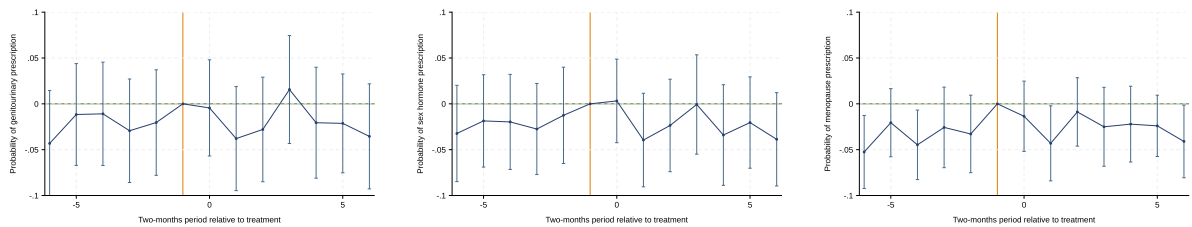


Figure A3: Event-study estimates for genitourinary medications (left), hormone medications (centre), and menopausal hormone therapy (right) among all women in the treated and control towns. The horizontal axis measures two-month periods relative to the first RWGPS visit; the reference period is the two-month period before entry.

B Town-Day Analysis and Timing Validation

B.1 Town-Day (Reduced-Form) Analysis

Empirical Strategy

This appendix presents reduced-form corroboration of the within-patient results at the town-day level. This specification asks only whether town-wide outcomes shift on visit days, without requiring observation of which provider any individual patient saw:

$$Y_{td} = \alpha + \beta \text{VisitDay}_{td} + \gamma_{t \times ym} + \delta_{t \times dow} + \varepsilon_{td}, \quad (4)$$

where Y_{td} denotes the mean outcome in town t on date d , VisitDay_{td} equals one if the visiting female GP held a clinic that day, $\gamma_{t \times ym}$ denotes town-by-year-month fixed effects, and $\delta_{t \times dow}$ denotes town-by-day-of-week fixed effects. Standard errors are clustered at the town-by-month level.

Consultation Length

Table [B1](#) presents town-day consultation length results. The share of long consultations increases by 11 percentage points on visit days among all patients, and by 22 percentage points among women; the estimate is close to zero among men.

Table B1: Consultation Length: Town-Day Estimates

	(1)	(2)	(3)
	All patients	Women	Men
<i>Reduced Form (Town-Day)</i>			
Visit day	0.114*** (0.005)	0.218*** (0.008)	−0.002 (0.006)
Observations	18,664	16,224	16,004
R-squared	0.239	0.395	0.310
Baseline mean	0.19	0.20	0.18

Notes: Outcome is the share of consultations exceeding 20 minutes. Reduced-form estimates at the town-day level with town-by-year-month and town-by-day-of-week fixed effects; standard errors clustered at town-by-month level. Baseline mean calculated on non-visit days. Corroborates Table 6.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Prescription Outcomes

Table B2 presents town-day prescription results. The shares of prescription days that include a genitourinary medication or a hormone medication rise by approximately 4 percentage points on visit days. Cardiovascular prescribing also increases, but by a substantially smaller 1.3 percentage points, while mental health prescribing shows a small negative coefficient. Because the town-day specification captures aggregate changes in prescribing on visit days, it does not isolate care delivered by the female GP. In the within-patient design, which compares the same woman’s prescription days across providers with town-by-date fixed effects, cardiovascular and mental health prescribing show no statistically significant differences, while the differences for genitourinary and hormone medications are an order of magnitude larger.

Table B2: Prescription Outcomes by Category: Town-Day Estimates (Women Only)

	(1)	(2)	(3)	(4)
	Genitourinary	Hormone	Mental health	Cardiovascular
	medications	medications	medications	medications
<i>Reduced Form (Town-Day)</i>				
Visit day	0.041***	0.039***	−0.005*	0.013**
	(0.004)	(0.003)	(0.003)	(0.005)
Observations	20,889	20,889	20,889	20,889
R-squared	0.064	0.061	0.101	0.10

Notes: Sample restricted to female patients. Outcome is the share of prescription days in each therapeutic category at the town-day level. Reduced-form estimates with town-by-year-month and town-by-day-of-week fixed effects; standard errors clustered at town-by-month level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Cervical Screening

Table B3 presents the corresponding cervical screening result: screening increases by 1.9 percentage points on visit days, against a town-day baseline mean of 0.4 percent.

Table B3: Cervical Screening: Town-Day Estimates

Cervical screening	
<i>Reduced Form (Town-Day)</i>	
Visit day	0.019*** (0.001)
Observations	21,578
R-squared	0.08
Baseline mean	0.004

Notes: Sample restricted to women aged 25–74. Reduced-form estimates at the town-day level with town-by-year-month and town-by-day-of-week fixed effects; standard errors clustered at town-by-month level. The lower baseline mean relative to Table 4 reflects the town-day level of aggregation, which averages over many days with zero screenings. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B.2 Timing (Placebo-Date) Validation

Because the town-day design identifies effects from the timing of visits, I assess whether the estimates reflect the actual visit dates rather than broader temporal patterns. Shifting visit dates by ± 7 days yields estimates that are small relative to the true effects (Tables B4–B6). The largest consultation-length estimate, at 0.009, is marginally significant and equals eight percent of the corresponding true effect of 0.114. The prescription and screening estimates are small, of mixed sign, and statistically insignificant.

Table B4: Placebo Dates: Consultation Length (Town-Day)

	Placebo Dates	
	+7 days	−7 days
Coefficient	0.009*	0.0005
	(0.005)	(0.005)
True effect (Reduced Form)	0.114	
Observations	17,460	17,460
R-squared	0.221	0.220

Notes: Reduced-form estimates at the town-day level with town-by-year-month and town-by-day-of-week fixed effects, using placebo visit dates shifted by ± 7 days. The outcome is the share of long consultations. Standard errors clustered at the town-by-month level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B5: Placebo Dates: Prescription Outcomes (Women Only, Town-Day)

	Genitourinary medications		Hormone medications		Menopausal hormone therapy		Contraceptives	
	+7 days	−7 days	+7 days	−7 days	+7 days	−7 days	+7 days	−7 days
Coefficient	0.002	−0.003	0.002	−0.004	−0.0004	−0.002	0.001	−0.001
	(0.004)	(0.003)	(0.003)	(0.003)	(0.002)	(0.002)	(0.003)	(0.002)
Observations	19,691	19,691	19,691	19,691	19,691	19,691	19,691	19,691
R-squared	0.054	0.054	0.054	0.054	0.055	0.055	0.049	0.049

Notes: Sample restricted to female patients. Reduced-form estimates at the town-day level using placebo visit dates shifted by ± 7 days; specifications include town-by-year-month and town-by-day-of-week fixed effects, with standard errors clustered at the town-by-month level. Menopausal hormone therapy and contraceptive columns restrict to women aged 45 and over and to women under 45 respectively. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B6: Placebo Dates: Cervical Screening (Town-Day)

	Placebo Dates	
	+7 days	−7 days
Coefficient	0.0001	−0.0003
	(0.0006)	(0.0007)
True effect (Reduced Form)	0.019	
Observations	20,370	20,370
R-squared	0.05	0.05

Notes: Sample restricted to women aged 25–74. Reduced-form estimates using placebo visit dates shifted by ± 7 days; standard errors clustered at the town-by-month level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C Qualitative Evidence

C.1 Additional Qualitative Evidence

Patient Perspectives

Comfort, listening, and unhurried consultations

"A female GP who understands post menopausal women. You could take a list of items you wanted to discuss and she worked through each of them. Extremely thorough without feeling any sense of rushing. I felt listened to and could ask anything."

RWGPs client (2025 survey)

"The doctor spoke clear and slowly as I have a poor hearing -- she was also extremely thorough and sent me off for scan and CT after explaining to me the reasons."

RWGPs client (2025 survey)

"It's been really good for young girls who want somebody that they feel comfortable with -- and the mums feel comfortable too. Our community was crying out for a service like this."

RWGPs client, community member (RFDS Victoria Annual Report 2018/19)

Selective use for gender-sensitive care

"Be able to talk to a female doctor. This is something we are lacking in the country."

RWGPs client (2023 survey)

"I was encouraged by the female GP to have a test that I had been avoiding for years."

RWGPs client (2023 survey)

"Great for my young daughter to see a female GP regarding her health."

RWGPs client (2025 survey)

"Offered locally. It's hard to get appointments in larger centres and it is difficult to me to get away during this drought."

RWGPs client (2025 survey)

Figure C1: Patient voices, Rural Women's GP Service (anonymised). Quotes drawn from client feedback surveys and RFDS Victoria annual reports.

Provider and Host-Clinic Perspectives

Anchor quote: an RWGPS GP describes the selectivity pattern verbally

"Women come to me for women's health issues like pap tests or menopause management, but as we build a rapport they come to me for regular care, such as blood pressure checks or counselling."

RWGPS GP, quoted in RFDS Victoria Annual Report 2016/17 (name withheld).

<i>"There are many reasons that may be quite obvious as to why a woman would like to see a female GP, but there are also times when a man may feel more comfortable talking to a woman, especially regarding mental health and social issues."</i> RWGPS GP, RFDS Victoria Annual Report 2015/16	<i>"There are also issues of privacy and confidentiality in a small town... It helps to have a choice to have another person that goes into the town and goes back to the city and they don't run into them in the supermarket."</i> RWGPS GP, RFDS Victoria Annual Report 2015/16
<i>"They often ask for my knowledge and also a bit of peer support."</i> RWGPS GP, RFDS Victoria semi-structured interview, 2025	<i>"It does lighten the load for the team. It's good to have an extra set of hands. It disperses the load and [the host site team] can deal more with their acute or hospital work."</i> Host clinic stakeholder, RFDS Victoria 2023/2025

Figure C2: Provider and host-clinic stakeholder voices (anonymised).